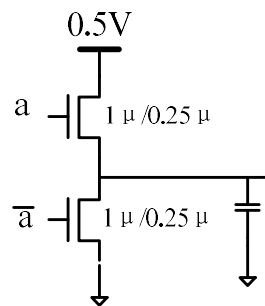

2021-2022 《数字 VLSI》第二次课程作业

一、请分别画出实现 $F = \bar{A} + \bar{B}\bar{C} + \bar{D}$ 功能的复合静态互补CMOS逻辑门的晶体管电路并标注每个晶体管的尺寸系数, 计算它的本征延时、所有输入端的逻辑努力。

二、下图所示电路是一个低摆幅总线的驱动器, 已知 NMOS 的 $V_T=0.43V$, $V_{DSAT}=0.63V$, $k'=115 \times 10^{-6} A/V^2$, 沟道尺寸如图所示, 输入信号摆幅等于 $2.5V$, 驱动器电源电压是 $0.5V$, 总负载电容为 $1pF$, 请计算驱动器的上升传播延时和下降传播延时 (忽略体效应和沟道长度调制效应, 晶体管等效导通电阻取初态值和末态值的平均值)。

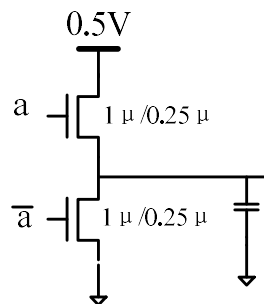


三、一个 3 输入 NAND 门驱动一个电容, 等效扇出为 48, 已知自载系数为 1, 请写出该逻辑门的归一化延时。

2021-2022 《digital VLSI》

HOMEWORK #2

- I. Please draw the schematic of the static CMOS logic gate that implements the Boolean function $F = \bar{A} + \bar{B}\bar{C} + \bar{D}$ and mark size coefficient of each transistor, calculate the normalized eigen delay τ_p and logical efforts at each input of this logic gate. (Symmetrical inverter $W_p:W_n=2:1$)
- II. There is a low-swing bus driver as follow. The NMOS FET's parameters have been known ($V_T=0.43V$, $V_{DSAT}=0.63V$, $k'=115 \times 10^{-6} A/V^2$), the channel size is also shown in the figure. The swing of input signal is $2.5V$, the supply voltage of this driver is $0.5V$, and the total load capacitance is $1 pF$. Please calculate the rise and the fall propagation delay of this driver. (Ignore the body effect and the channel length modulation effect. The transistor equivalent turn-on resistance takes the average value of the initial state value and the final state value.)



- III. Consider a 3-input NAND gate (self-load factor = 1). Knowing that the electrical effort from its input to its output pin equals 48, please give the normalized delay of this logic gate.